

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S1 (S, FE) S2 (S, FE) Examination December 2024 (2019 Scheme)

Course Code: EST102**Course Name: PROGRAMMING IN C**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
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| 1 | Differentiate between high level and low level languages. | (3) |
| 2 | Write an algorithm to find the largest of three numbers. | (3) |
| 3 | Write the equivalent while loop for the following C code.
int i;
for (i=10;i<0;--i)
print ("%d",i); | (3) |
| 4 | Differentiate between assignment and equality operators? | (3) |
| 5 | Explain the different methods to declare and initialize a single dimensional character array with example. | (3) |
| 6 | Write the C code to find the length of a string without using built in function. | (3) |
| 7 | Compare structure and union in C. | (3) |
| 8 | Write the syntax for defining a function in C. Explain with example | (3) |
| 9 | Explain null pointer in C. | (3) |
| 10 | Write the purpose of file mode? Mention any four file modes. | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

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|----|--|-----|
| 11 | a) Explain the different types of memory used in a computer. | (8) |
| | b) Write an algorithm to find reverse of an n digit number. | (6) |

OR

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| 12 | a) Explain the functional units of a computer with the help of a diagram. | (8) |
| | b) Draw a flowchart to check whether the given year is leap year or not. | (6) |

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Module II

- 13 a) Write a C Program to find the sum of odd and even numbers separately between a range. (6)
b) Explain the difference between while loop and do while loop with example for each. (8)

OR

- 14 a) Explain the different types of operators in C with examples (8)
b) Write a C program to generate the Fibonacci series up to 50. (6)

Module III

- 15 a) Read a string (ending with a \$ symbol), store it in an array and count the number of vowels and consonants in it. (6)
b) Explain any four string handling functions with example for each. (8)

OR

- 16 a) Write a C program to sort a set of names in alphabetical order. (9)
b) Write a C program to search a number in an array of n integer numbers. (5)

Module IV

- 17 a) Write a C program to find the factorial of a number using function. (7)
b) Write a C program to check whether a number is prime or not using function. (7)

OR

- 18 a) Explain the different types of storage classes in C with examples for each. (6)
b) Declare a structure namely Student to store the details (roll number, name, mark_for_C) of a student. Then, write a program in C to find the average mark obtained by the students in a class for the subject Programming in C (using the field mark_for_C). Use array of structures to store the required data. (8)

Module V

- 19 a) Create a file and perform the following (9)
i) Write data to the file
ii) Read the data in a given file & display the file content on console
iii) append new data and display on console
b) Explain the pass by reference mechanism with an example (5)

OR

- 20 a) Write a C program to swap two numbers using pointers. (7)
b) Explain any three file handling functions in C. (7)
